The Central Distribution: A Noninformative Prior for Feature Selection

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1 Introduction

We begin by revisiting a fundamental decision-making problem, and how it is treated within the framework of Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The loss associated with each action depends on the unknown state of the environment and can be quantified by a real number. Specifically, let a denote an action, and x represent the unknown environment state. The loss function $\ell(a, x)$ quantifies the loss for each action.

Furthermore, we assume that an evidence e of this unknown environment state exists, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find an optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence: $S(A \mid E)$.

According to Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg\min_S \sum_a \sum_{x,e} \ell(a, x) S(a \mid e) P^*(x, e),$$

where P^* is the joint probability distribution of the environment state and the evidence.

It can be shown that there always exists a deterministic optimal strategy. That is, we can consider a strategy as a function of the evidence, rather than as a conditional distribution:

$$S_{\text{opt}} = \arg\min_S \sum_{x,e} \ell(S(e), x) P^*(x, e).$$

A strategy that, for each evidence e, selects the action minimizing the expected loss conditioned on e, is optimal:

$$\begin{aligned} S_{\text{opt}}(e) &= \arg\min_a \mathbb{E}_{P^*}\left[\ell(a,x) \mid e\right] \\ &= \arg\min_a \sum_x \ell(a,x) P^*(x \mid e)\cdot \end{aligned}$$

This shows that the optimal actions are determined solely based on $P^*(X|E)$. Note that the value of the expected loss depends on $P^*(E)$, but the actions that construct the optimal strategy are not affected by $P^*(E)$. Only for suboptimal strategies, $P^*(E)$ plays a role and determines the importance of suboptimal actions.

To identify the optimal strategy, we rely on P^*. But how can we obtain this distribution? Is there an inherent, objective distribution within the environment that we must discover?

Answering this question is not straightforward. From a subjective perspective on probability, the answer is negative. There is no such objective distribution, and P^* exists only in our minds. It serves as a tool that helps us regulate our reasoning—an intermediate measure created within the Bayesian decision-making framework to ensure we stay consistent with our assumptions and common sense principles.

Considering the subjective viewpoint, the determination of P^* relies on one's prior knowledge and assumptions. When we have no assumptions and no prior knowledge, the set of candidate probabilities for P^* simply consists of all possible probability measures. As we make assumptions or obtain some prior knowledge, those probability measures that do not align with our assumptions or knowledge are eliminated from the set. Consequently, the set of acceptable probability measures becomes more refined. We will refer to this refined set as a probabilistic model:

Definition 1.1 (Probabilistic Model)**.** We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of acceptable probability measures on $\mathcal{S}$.

To narrow the set of acceptable probabilities, one of the most commonly used assumptions in the statistical modeling, is the exchangeability assumption. We say a sequence of random variables $X_1, X_2, \ldots$ is exchangeable, if the ordering of variables does not provide any useful information. In other words, for an exchangeable sequence of random variables, the joint probability is the same for any permutation of variables. This property holds true for every subset of the sequence.

For example, if each X_i represents a binary variable, the joint probability depends solely on the total count of 1s and 0s, and the specific order in which these variables appear becomes irrelevant.

When a sequence of random variables is exchangeable, de Finetti's theorem states that their joint probability has a special representation:

$$P(x_1, x_2, \ldots) = \int \prod_i \pi(x_i \mid \theta) p(\theta) d\theta.$$

This means, exchangeable observations are conditionally independent relative to some latent parameter θ.

Moreover, the theorem states that if each X_i is discrete, the distribution $\pi(x_i \mid \theta)$ corresponds to the multinomial distribution.

Definition 1.2 (Parametric Model)**.** Let Θ be a real vector. A *parametric model* is a probabilistic model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability distribution p, such that for every $s \in \mathcal{S}$

$$P(s) = \int \pi(s \mid \theta) p(\theta) d\theta, \tag{1}$$

where π is a fixed, known probability distribution. We refer to Θ as the *model parameter* and π as the *likelihood* distribution. We say that P is the probability measure *induced* by p.

In our decision making problem, suppose that the available evidence E, consists of a sequence of previous experiences of the environment. If we assume that our environment remains unchanged over time so that this sequence is exchangeable, to fully determine P^*, we only need to determine the prior distribution of the latent parameter $p(\theta)$.

Clearly, one approach to selecting a suitable prior distribution $p(\theta)$ is to rely on our prior knowledge about the environment. But, what if such knowledge is unavailable? This is a crucial question. In Bayesian statistics, this is where noninformative priors come into play. These priors are typically chosen to be uniform or to represent a state of ignorance about the parameters.

There is also another consideration that could play a role in choosing an appropriate prior distribution. Suppose that in our initial decision making problem, we decide to use a different evidence E' instead of E. Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can easily answer this question: To compare any two strategies, we can simply compare their expected loss. The strategy with the lower expected loss is the better one.

Note that, in reality, we cannot freely choose any random variable as our evidence. We can only choose evidence that is a deterministic function of the evidence provided by the environment, allowing us to determine its value after observing the original evidence. Utilizing a strategy that uses a simpler evidence, resembles feature selection in machine learning. For instance, in a learning task where our evidence is an image, we might substitute the image with the first three coefficients of its Fourier transform. Naturally, these coefficients are a function of the original evidence.

Interestingly, if we assume $E' = f(E)$, it turns out that, according to our Bayesian framework, the optimal strategy for E is always better than any strategy that uses E'. Thus, at first glance, it appears that Bayesian decision theory does not favor feature selection.

From a certain perspective, it could be argued that adopting a strategy which uses simpler evidence may lead to ignoring some parts of useful available information, and therefore is not justified by Bayesian decision theory. However, in practice we rather to use feature selection because without it the original learning task requires impractical large datasets.

The key point here is that the space of strategies using E' is a subset of the strategies that use E. When we are searching for the best strategy based on E, we are implicitly evaluating all simpler forms as well. We believe this is why the Bayesian framework never favors the simpler evidence E'. Plausibly, if we choose an appropriate prior distribution, when we are finding the optimal strategy, feature selection should be performed automatically within the Bayesian framework. However, many commonly used non-informative priors, such as the flat prior, do not exhibit feature selection characteristics.

Throughout this discussion, we will explore a systematic approach for finding a noninformative prior that also exhibit feature selection characteristics.

2 Central Distribution

When we lack prior knowledge of an environment, it is still not difficult to identify some distributions that are not suitable choices for $p(\theta)$. For instance, any distribution that assigns zero probability to certain values of θ is not a good choice. If we use such a prior, evidence can never convince us that the value of θ lies within those excluded values.

We must point out that, in this context, we cannot objectively claim that there exists an underlying process within the environment for generating different values of θ, which does not consider zero probability for any value of θ. Our argument is subjective here. We consider some priors to be unsafe because they have some characteristics that appear to be problematic: in the obtained posterior, some values of the parameter are excluded regardless of the evidence.

We can generalize this idea and use a similar subjective approach to determine a somewhat "safe" prior distribution. Recall that our probabilistic model for the environment consists of a set of acceptable probability measures $\mathcal{P}$. Suppose that we choose a probability measure $C \in \mathcal{P}$, and determine the optimal Bayesian strategy in C. If the expected loss of this strategy in every other probability measure $P \in \mathcal{P}$ is close to the optimal strategy, we can say that using C is safe, in the sense that the optimal strategy obtained using C, is a near optimal strategy in any other acceptable probability measure in our model.

The probability measure that we choose, represents our beliefs about the environment. It can be argued that this choice must depend solely on the environment. As long as the environment is the same, we must not change this choice. For example, if our loss function changes, or if the evidence that we get from the environment becomes more or less informative, none of these should change our beliefs about the environment, and therefore our chosen probability measure. We will use this principle several times in this discussion and refer to it as the *principle of objectivity*.

2.1 Divergence

Unfortunately, the expected loss of a strategy is a functional of the loss function. Therefore, based on the principle of objectivity, we should not directly use the expected loss of the optimal Bayesian strategy for evaluating probability measures. We need a measure that quantifies the similarity between two probability measures while it ignores the specific loss function we are using.

Suppose that for an environment, we have decided on a probabilistic model $\mathcal{M}$, and without any particular prior knowledge, we have chosen a probability measure P from our model. In particular, we are interested in the distribution of a random variable X.

Imagine that we ask an expert—a person familiar with the environment—to evaluate our probability measure P. Presumably, that expert has his own beliefs about the environment, and thinks that we should, instead of P, use a different probability measure Q. When evaluating P, he would try to measure the loss of using the incorrect distribution $P(X)$, while he believes the correct distribution is $Q(X)$.

Let us assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left(Q_X \parallel P_X\right)$. It seems reasonable, as P gets closer to Q

this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}\left(Q_X \parallel P_X\right)$ effectively is a measure of the divergence of $P(X)$ from $Q(X)$, when Q is the believed true probability measure.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback–Leibler (KL) divergence.

Alternatively, by adopting a more minimal set of properties, we can prove that if we accept Shannon's entropy as a measure of the uncertainty of a distribution, we should also consider KL divergence as the appropriate method for measuring divergence between distributions.

Theorem 1. *Let X be a discrete random variable. Suppose that $\mathbb{D}\left(Q_X \parallel P_X\right)$ is a divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

Consistency with entropy *Let Ω be a continuous random variable, and Q be a probability measure. If $U(\Omega)$ is the uniform distribution with the same support as $Q(\Omega)$, then we have*

$$\mathbb{D}\left(Q_\Omega \parallel U_\Omega\right) = \mathbb{H}\left[U_\Omega\right] - \mathbb{H}\left[Q_\Omega\right],$$

where $\mathbb{H}$ denotes Shannon's entropy.

Additivity *For any two discrete random variables X, Y, and probability measures P, Q*

$$\mathbb{D}\left(Q_{X,Y} \parallel P_{X,Y}\right) = \mathbb{D}\left(Q_X \parallel P_X\right) + \sum_x \mathbb{D}\left(Q_{Y|x} \parallel P_{Y|x}\right) Q(x)\cdot$$

If $\mathbb{D}$ satisfies these properties, then it is the Kullback-Leibler divergence.

Proof. Let $P(X)$ and $Q(X)$ be two arbitrary distributions for a discrete random variable X. We define a new continuous random variable Ω and let

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\Omega \mid X)$ and $P(\Omega, X)$ are uniform distributions, and $\mathbb{H}\left[P_{\Omega|x}\right] = \ln P(x)$, $\mathbb{H}\left[P_{\Omega,X}\right] = 0$.

We define $Q(\Omega \mid x)$ to be an arbitrary distribution that has the same support as $P(\Omega \mid x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned}
\mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) &= \mathbb{H}\left[P_{\Omega,X}\right] - \mathbb{H}\left[Q_{\Omega,X}\right] \\
&= \sum_x \int q(\omega,x) \ln q(\omega,x) d\omega \\
&= \sum_x Q(x) \int q(\theta \mid x) \ln q(\omega,x) d\omega,
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) &= \mathbb{H}\left[P_{\Omega|x}\right] - \mathbb{H}\left[Q_{\Omega|x}\right] \\
&= \ln P(x) + \int q(\theta \mid x) \ln q(\theta \mid x) d\omega \cdot
\end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned}
\mathbb{D}\left(Q_X \parallel P_X\right) &= \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) - \sum_x \mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) Q(x) \\
&= \sum_x Q(x) \left(\int q(\theta \mid x) \ln \frac{q(\omega,x)}{q(\theta \mid x)} d\omega - \ln P(x) \right) \\
&= \sum_x Q(x) \left(\ln Q(x) \int q(\theta \mid x) d\omega - \ln P(x) \right) \\
&= \sum_x Q(x) \ln \frac{Q(x)}{P(x)} \cdot
\end{aligned}$$

□

In this theorem, the first property simply states that the entropy of a distribution should be related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X,Y)$ from $Q(X,Y)$ is a measure of the generic loss of using $P(X,Y)$ when we believe $Q(X,Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with the probability $Q(x)$, it would also include the loss of using $P(Y \mid X = x)$. This property indicates that for combining these individual terms we should use addition.

Thus, if Q is the probability measure that represents our beliefs, we can use $\mathbb{D}\left(Q_X \parallel P_X\right)$ to evaluate any other probability measure P, with respect to a random variable X. In other words, we evaluate $P(X)$ by $Q(X)$.

If we have two random variables, X and Y, there would be two distinct cases that should not be confused: We may want to evaluate $P(X,Y)$, or we may want to evaluate $P(X)$ and $P(Y)$ separately. Evaluating the joint distribution $P(X,Y)$ is straightforward and can be done by defining a new random variable $Z = (X,Y)$.

The second case, however, is more challenging and the correct approach is not immediately clear. Here, we may use a similar argument to what we used for the additive property: Suppose that we are evaluating the loss of using $P(X,Y)$, when X and Y are independent. If X happens first, we will use the marginal $P(X)$. Therefore, the loss of using $P(X,Y)$ includes the loss of using $P(X)$. Then, when Y happens, since X and Y are independent, we will be using the marginal $P(Y)$, and the loss of using $P(X,Y)$ also includes the loss of using $P(Y)$. Therefore, it appears that separately evaluating the marginal distributions $P(X)$ and $P(Y)$, should be the same as evaluating $P(X,Y)$ when X and Y are independent. Hence, we may use

$$\mathbb{D}\left(Q_X \parallel P_X\right) + \mathbb{D}\left(Q_Y \parallel P_Y\right).$$

Separate evaluation of marginals is essential for the divergence measure that we will define shortly.

The KL divergence provides a method for measuring the agreement of a model about a specific distribution. This agreement can be called convergence.

Definition 2.1 (ϵ-convergence)**.** In a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, let X and E be discrete random variables. We say $X|E$ (X conditioned on E) *ϵ-converges by* $P \in \mathcal{P}$, if for any probability measure $Q \in \mathcal{P}$ we have:

$$\mathbb{E}_Q\left[\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right)\right] < \epsilon,$$

where $\mathbb{E}_Q\left[\,\cdot\,\right]$ denotes expectation relative to Q and $\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right)$ is the Kullback-Leibler divergence of $P(X \mid e)$ from $Q(X \mid e)$:

$$\mathbb{D}\left(Q_{X|e} \parallel P_{X|e}\right) = \sum_x Q(x \mid e) \ln \frac{Q(x \mid e)}{P(x \mid e)},$$

with the convention that $\ln(0) \equiv 0$.

ϵ-convergence is closely related to the concept of Bayesian convergence, where the posterior converges to the same distribution regardless of the prior distribution.

As an example, consider a scenario where we are trying to predict the outcome of a four-sided dice. This dice is unfair, with each face marked by a binary number: 00, 01, 10, and 11. Before making our prediction, we can roll the dice multiple times, with each roll being independent.

If we let E be a vector representing the first 10 dice rolls and X be the outcome of the next roll, $X|E$ ϵ-converges by P_u for $\epsilon \geq 0.109$, where P_u is the probability measure induced by considering a uniform prior for the multinomial parameters. If E represents the first 20 outcomes, then we can choose any $\epsilon \geq 0.073$.

Here, the selected prior for the multinomial parameters affects the minimum value of ϵ. In other words, different probability measures in our model have different ϵ-convergence properties. The minimum ϵ gives us an assessment of the safety of a probability measure, with respect to a particular $X|E$ conditional. If this value is sufficiently small, any other probability measure in the model agrees that our distribution for $X|E$ is safe, in the sense that it has a small expected divergence.

When we are interested in the distribution of several random variables, we may consider the sum of their KL divergences, as we discussed earlier, to construct a measure that accounts for various random variables.

Definition 2.2 (Query Set)**.** A *query set* is a set of pairs of discrete random variables, all defined on the same sample space.

Definition 2.3 (Divergence Measure)**.** Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$, be a query set in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure of $P \in \mathcal{P}$ for $\mathcal{X}$* as follows:

$$\begin{aligned}\mathbb{D}_{\mathcal{X}}\left[P\right] &= \sup_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right], \\ &= \sup_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(X_i = x_i, E_i = e_i) \ln \frac{Q(X_i = x_i \mid E_i = e_i)}{P(X_i = x_i \mid E_i = e_i)}.\end{aligned} \tag{2}$$

Proposition. $X|E$ ϵ-converges by P, if and only if

$$\mathbb{D}_{\{(X,E)\}}\left[P\right] < \epsilon.$$

Note that we usually simplify our notation and instead of writing the full assignment to a random variable, we only write the assigned value. For example, instead of $Q(X_i = x_i, E_i = e_i)$ we write: $Q(x_i, e_i)$.

For a parametric model (as defined in Definition 1.2), it seems reasonable to define a conditional form of our divergence measure, conditioned on the value of the parameter.

To do so, in Equation 2, we replace normal expectation with conditional expectation. There will be no longer a need to maximize over different distributions, since in a parametric model the parameter fully determines the distribution of the sample space.

Definition 2.4 (Conditional Divergence Measure)**.** Assume $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ is a query set. We define the *conditional divergence measure of $P \in \mathcal{P}$ for $\mathcal{X}$ given θ* as follows:

$$\mathbb{D}_\mathcal{X}\left[P \mid \theta\right] = \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \, \theta \right] \tag{3}$$

$$= \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi\left(x_i \mid e_i, \theta\right)}{P\left(x_i \mid e_i\right)} \cdot \tag{4}$$

Proposition. For the conditional divergence we have:

- $\mathbb{D}_\mathcal{X}\left[P \mid \theta\right] \geq 0$
- $\mathbb{D}_\mathcal{X}\left[P\right] = \sup_{Q \in \mathcal{P}} \int \left(\mathbb{D}_\mathcal{X}\left[P \mid \theta\right] - \mathbb{D}_\mathcal{X}\left[Q \mid \theta\right]\right) q(\theta) d\theta$

Subjectively, the probability measure that has the lowest divergence can be considered the "safest" choice in a model. It is safe in the sense that it has the best possible performance in the worst case, and more importantly, when convergence is possible in a model, if we use this probability measure, we are guaranteed to have a convergent distribution for our query set.

Definition 2.5 (Central Distribution)**.** In a model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, we define the *central distribution* for a query set $\mathcal{X}$, to be a probability measure $C_\mathcal{X} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_\mathcal{X} = \arg\min_{P \in \mathcal{P}} \mathbb{D}_\mathcal{X}\left[P\right] \cdot$$

In this definition, the query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ plays a crucial role. Specifically, $\mathcal{X}$ defines the set of random variables in which we are interested. Each X_i represents a random variable whose distribution is important to us, and each E_i corresponds to the evidence that is available for that variable.

However, if we choose our query set based on our specific experimental setup, it would violate the principle of objectivity. To avoid that, we have to use the same query set for any possible problem that could be defined within an environment.

One natural way to do this is to let the query set include all potentially relevant conditional pairs. For example, in our dice problem, we might let each X_i be the outcome of the i-th dice roll, and define each E_i as a vector of the previous dice outcomes: $E_i = (X_1, X_2, \dots, X_{i-1})$. We may assume that our query set contains n pairs. The choice of n should be based on our assumptions about the complexity of the problem. It shows where we think the convergence becomes trivial and, if necessary, could approach infinity.

Now let us calculate the conditional divergence measure (Definition 2.4) for this query set:

$$\begin{aligned}
\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] = & \sum_{x_1} \pi(x_1 \mid \theta) \ln \frac{\pi(x_1 \mid \theta)}{P(x_1)} \\
& + \sum_{x_2, x_1} \pi(x_2, x_1 \mid \theta) \ln \frac{\pi\left(x_2 \mid x_1, \theta\right)}{P\left(x_2 \mid x_1\right)} \\
& + \cdots \\
& + \sum_{x_n, x_1, \dots, x_{n-1}} \pi(x_n, x_1, \dots, x_{n-1} \mid \theta) \ln \frac{\pi\left(x_n \mid x_1, \dots, x_{n-1}, \theta\right)}{P\left(x_n \mid x_1, \dots, x_{n-1}\right)}.
\end{aligned}$$

After factoring out $\pi(x_1, \dots, x_n \mid \theta)$ from every term:

$$\sum_{x_1, \dots, x_n} \pi(x_1, \dots, x_n \mid \theta) \ln \frac{\pi(x_1 \mid \theta)\pi(x_2 \mid x_1, \theta) \dots \pi(x_n \mid x_1, \dots, x_{n-1}, \theta)}{P(x_1)P(x_2 \mid x_1) \dots P(x_n \mid x_1, \dots, x_{n-1})},$$

we use the chain rule for conditional probability:

$$\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] = \sum_{x_1, \dots, x_n} \pi(x_1, \dots, x_n \mid \theta) \ln \frac{\pi(x_1, \dots, x_n \mid \theta)}{P(x_1, \dots, x_n)}. \tag{5}$$

This means that we can alternatively define our query set as $\mathcal{X} = \{(X, E)\}$, where X represents the vector of n dice rolls: $X = (X_1, X_2, \dots, X_n)$, and E is a dummy certain event with $P(E) = 1$. This query set essentially corresponds to the entire sample set for n dice rolls. We usually simplify our notation and omit dummy events and write: $\mathcal{X} = \{X\}$.

The fact that our query set includes the entire sample set does not necessarily mean its central distribution has good convergence properties for every random variable. In our dice example, suppose we are interested in predicting the next dice outcome based on the number of previous odd and even outcomes. Let Y_i represent the least significant bit of the i-th dice roll, indicating whether the number is odd or even. If we want our central distribution to exhibit good convergence properties for $P(X_i \mid Y_1, Y_2, \dots, Y_{i-1})$, we need to explicitly include these conditional pairs in our query set, in the same way we did for $X_i|X_1, X_2, \ldots, X_{i-1}$.

Intuitively, it seems that in any noninformative distribution we should have

$$P(X_i \mid Y_1, Y_2, \ldots, Y_{i-1}) = \frac{1}{2} P(Y_i \mid Y_1, Y_2, \ldots, Y_{i-1}).$$

For example, consider $P(X_4 = 01 \mid 0, 1, 1)$, which represents the probability of observing 01 on the 4th roll, given that we have previously seen two odd numbers and one even number. It is reasonable to expect that this probability equals $P(X_4 = 11 \mid 0, 1, 1)$. On the other hand,

$$P(Y_4 = 1 \mid 0, 1, 1) = P(X_4 = 01 \mid 0, 1, 1) + P(X_4 = 11 \mid 0, 1, 1),$$

which implies

$$P(X_4 = 01 \mid 0, 1, 1) = P(X_4 = 11 \mid 0, 1, 1) = \frac{1}{2} P(Y_4 = 1 \mid 0, 1, 1).$$

This suggests that adding $Y_i \mid Y_1, Y_2, \ldots, Y_{i-1}$ conditionals to our query set, would have the same effect as adding $X_i \mid Y_1, Y_2, \ldots, Y_{i-1}$ conditionals.

If we add conditionals of the form $Y_i \mid Y_1, Y_2, \ldots, Y_{i-1}$ to our query set, we will get $\mathcal{X} = \{X, Y\}$, where X represents the sample space of n rolls of a four sided dice, and Y represents the sample space of n rolls of a two-sided odd-even dice.

The central distribution for this new query set shows some interesting properties. In this distribution, for smaller values of i, $P(X_i \mid X_1, X_2, \ldots, X_{i-1})$ tends to get somewhat combined with $P(X_i \mid Y_1, Y_2, \ldots, Y_{i-1})$. For example when $i = 2$, if the single observed dice roll is 01, the posterior probabilities of $00, 10$ and 11 are not equal, and $P(11 \mid 01) = 0.158$, which is slightly higher than $P(00 \mid 01) = P(10 \mid 01) = 0.133$. We can say that the central distribution identifies that 01 and 11 are both odd numbers.

2.2 Parametric Models

In this section, we explore the process of deriving central distributions for parametric models. Our definition of parametric models is quite broad, entailing any model whose set of probability measures can be indexed by a finite-dimensional real vector. However, to establish a structured framework for identifying central distributions, we must refine this definition to include only models where the set of probability measures is isomorphic to a convex set.

Definition 2.6 (Generic Parametric Model)**.** We say a parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is *generic*, if for any arbitrary probability distribution r, there is a probability measure $R \in \mathcal{P}$, such that for every $s \in \mathcal{S}$

$$R(s) = \int \pi(s \mid \theta) r(\theta) d\theta \cdot$$

In a parametric model, the maximum of divergence happens for a prior distribution that is concentrated at a single value of the parameter θ. That means, for a parametric model, which allows any high peak prior, instead of maximizing over the set of all acceptable distributions, we can simply maximize over the parameter space, using the conditional version of the divergence measure.

Lemma 1. *Assume that $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model. Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a query set in $\mathcal{M}_\Theta$. For every $P \in \mathcal{P}$ we have*

$$\mathbb{D}_\mathcal{X}[P] \leq \sup_{\theta \in \Theta} \mathbb{D}_\mathcal{X}[P \mid \theta] \cdot$$

Proof. Let R and P be two arbitrary probability measures in $\mathcal{M}_\Theta$. For any parameter value θ we have:

$$\mathbb{D}_\mathcal{X}[P \mid \theta] - \mathbb{D}_\mathcal{X}[R \mid \theta] = \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \cdot$$

The conditional divergence is nonnegative and we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \mathbb{D}_\mathcal{X}[P \mid \theta] \cdot$$

This inequality holds for every θ. Therefore, the supremum of the right-hand side w.r.t. θ is also greater than or equal to the supremum of the left-hand side:

$$\sup_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sup_{\theta \in \Theta} \mathbb{D}_\mathcal{X}[P \mid \theta] \cdot \tag{6}$$

Trivially, for any θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sup_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \cdot$$

$\mathcal{M}_\Theta$ is a parametric model and $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$, where $r(\theta)$ is a prior probability distribution. Since the above inequality holds for every value of θ and $r(\theta) \geq 0$, we can multiply both sides by $r(\theta)$, and integrate. Notice that the right hand side is not a function of θ and it remains unchanged because $\int r(\theta)d\theta = 1$:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \leq \sup_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \cdot$$

Now from Inequality 6, for *any* $P \in \mathcal{P}$ and $R \in \mathcal{P}$, it follows that

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \leq \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}} \left[P \mid \theta\right],$$

which proves the lemma. □

Theorem 2. *Suppose $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model, and let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a query set in $\mathcal{M}_\Theta$. If for every $\epsilon > 0$ and every value of the parameter θ, there exists a distribution $R \in \mathcal{P}$ such that*

$$\mathbb{D}_{\mathcal{X}} \left[P \mid \theta\right] - \mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] < \epsilon, \tag{7}$$

then for any $P \in \mathcal{P}$ we have:

$$\mathbb{D}_{\mathcal{X}} \left[P\right] = \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}} \left[P \mid \theta\right] \cdot \tag{8}$$

Proof. Assume, for the sake of contradiction, that

$$\mathbb{D}_{\mathcal{X}} \left[P\right] \neq \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}} \left[P \mid \theta\right] \cdot$$

Then, Lemma 1 implies $\mathbb{D}_{\mathcal{X}} \left[P\right] < \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}} \left[P \mid \theta\right]$. Supremum is the minimal upper bound of a set. That means $\mathbb{D}_{\mathcal{X}} \left[P\right]$ can not be an upper bound for the set of values of $\mathbb{D}_{\mathcal{X}} \left[P \mid \theta\right]$, and there should exist a θ^* such that

$$\mathbb{D}_{\mathcal{X}} \left[P \mid \theta^*\right] > \mathbb{D}_{\mathcal{X}} \left[P\right] \cdot$$

By the theorem assumption, for θ^* and $\epsilon = \mathbb{D}_{\mathcal{X}} \left[P \mid \theta^*\right] - \mathbb{D}_{\mathcal{X}} \left[P\right]$ there should be a distribution $R \in \mathcal{P}$ such that Equation 7 holds, and

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] > \mathbb{D}_{\mathcal{X}} \left[P\right] \cdot$$

On the other hand

$$\sup_{Q\in\mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right] \geq \mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right].$$

Notice that the left hand side is $\mathbb{D}_{\mathcal{X}}[P]$. This contradiction implies that the assumption can not hold. Hence,

$$\mathbb{D}_{\mathcal{X}}[P] = \sup_{\theta\in\Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta]. \qquad \square$$

Proposition. A generic parametric model satisfies the conditions of Theorem 2.

The central distribution is essentially the solution to a minimax game. By relaxing some constraints, this game can be converted into a convex-concave game. It turns out that the solution to the relaxed problem always satisfies the constraints of the original problem, and is feasible. Since the solution to a convex-concave game corresponds to a saddle point, by defining the appropriate function, we can characterize the central distribution using standard convex optimization techniques.

Definition 2.7 (Saddle Point)**.** Consider a function $f : \mathcal{W} \times \mathcal{Z} \to \mathbb{R}$. We refer to a pair (w^*, z^*) as a saddle point for f, if we have

$$f(w^*, z) \leq f(w^*, z^*) \leq f(w, z^*),$$

for all $w \in \mathcal{W}$ and $z \in \mathcal{Z}$.

Lemma 2. *Let $\mathcal{M}_\Theta$ be a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a query set. Let f be the function defined by:*

$$f(\mathbf{P}_{\mathcal{X}}, r) = \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p_{x_i|e_i}}\right) r(\theta) d\theta,$$

where r is a probability distribution, and $\mathbf{P}_{\mathcal{X}}$ is a vector of real numbers with an entry $p_{x_i|e_i}$ for every i, x_i, and e_i, such that $\sum_{x_i} p_{x_i|e_i} = 1$. Additionally, let R denote the probability measure induced by r.

The point $(\mathbf{P}^*_{\mathcal{X}}, r^*)$ is a saddle point for f, if and only if there is a constant λ and a function μ, such that for any i, x_i, e_i, and θ the following conditions *hold:*

$$\sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p^*_{x_i|e_i}} - \mu(\theta) = \lambda,$$

$$p^*_{x_i|e_i} = R^*(x_i \mid e_i), \tag{9}$$

$$\int r^*(\theta) d\theta = 1,$$

$$r^*(\theta) \geq 0, \tag{10}$$

$$\mu(\theta) \geq 0,$$

$$\mu(\theta) r^*(\theta) = 0\cdot \tag{11}$$

Furthermore, a point $(\mathbf{Q}_{\mathcal{X}}, \tilde{r})$ *minimizes* $f(\mathbf{P}_{\mathcal{X}}, \tilde{r})$*, with respect to* $\mathbf{P}_{\mathcal{X}}$*, if and only if it satisfies*

$$q_{x_i|e_i} = \tilde{R}(x_i \mid e_i), \tag{12}$$

for all i*,* x_i*, and* e_i*.*

Proof. We define

$$\hat{f}(\mathbf{P}_{\mathcal{X}}, r) = \int \left(\sum_i \sum_{x_i,e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p_{x_i|e_i}} \right) r(\theta) d\theta,$$

where r is a real function of the model parameter θ, and $\mathbf{P}_{\mathcal{X}}$ is a vector of real numbers with an entry $p_{x_i|e_i}$ for every i, x_i, e_i.

Let $(\mathbf{P}^*_{\mathcal{X}}, r^*)$ be a simultaneous solution to the following two optimization problems:

$$\begin{array}{lll} \text{minimize} & \hat{f}(\mathbf{P}_{\mathcal{X}}, r) & \text{w.r.t. } \mathbf{P}_{\mathcal{X}} \\ \text{subject to} & \sum_{x_i} p_{x_i|e_i} = 1, & \text{for all } i, e_i \\ & & \\ \text{maximize} & \hat{f}(\mathbf{P}_{\mathcal{X}}, r) & \text{w.r.t. } r \\ \text{subject to} & r(\theta) \geq 0, & \text{for all } \theta \\ & \int r(\theta) d\theta = 1 & \end{array} \tag{13}$$

Clearly, if $(\mathbf{P}^*_{\mathcal{X}}, r^*)$ exists, it will be a saddle point for f.

Both of these optimization problems are convex [4] and differentiable. To establish this, consider that all the constraint functions are affine, and therefore are convex and differentiable. Note that for the constraint $r(\theta) \geq 0$, for each θ, we can define an affine constraint function of the form

$$\int \delta(\theta - t) r(t) dt \geq 0,$$

where δ is the Dirac delta function.

Additionally, $\hat{f}$ is differentiable function with respect to both r and $\mathbf{P}_{\mathcal{X}}$. For any fixed $\mathbf{P}_{\mathcal{X}}$, $\hat{f}(\mathbf{P}_{\mathcal{X}}, \cdot)$ is an affine function and is concave. For a fixed solution to the second optimization problem denoted by $\tilde{r}$, we can write

$$\hat{f}(\mathbf{P}_{\mathcal{X}}, \tilde{r}) = c - \sum_{i} \sum_{x_i, e_i} \tilde{R}(x_i, e_i) \ln p_{x_i|e_i}, \tag{14}$$

where c and $\tilde{R}(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) \tilde{r}(\theta) d\theta$ are not functions of $\mathbf{P}_{\mathcal{X}}$. Since $\tilde{r}$ is a solution to the second optimization problem, it satisfies the constraints of the second problem and is a distribution. Therefore, for all i, x_i, e_i we have $\tilde{R}(x_i, e_i) \geq 0$, and $\hat{f}(\cdot, \tilde{r})$ is the negative of a linear combination of concave $\ln p_{x_i|e_i}$ functions with nonnegative coefficients. This function is convex.

Thus, both optimization problems are convex and differentiable, with affine constraints. The constraints simply restrict the feasible set to the set of probability distributions, and it is always possible to find feasible points that meet all the constraints. Therefore, Slater's condition is satisfied, and the KKT conditions provide necessary and sufficient conditions for optimality [4]. In other words, If a point $(\mathbf{P}^*_{\mathcal{X}}, r^*)$ satisfies the KKT conditions for both problems, it will be a saddle point for f, and vice versa. The following KKT conditions are obtained for every θ, i, x_i, and e_i:

$$\int \pi(x_i, e_i \mid \theta) r^*(\theta) d\theta = \lambda(e_i) p^*_{x_i|e_i}, \tag{15}$$

$$\sum_{x_i} p^*_{x_i|e_i} = 1, \tag{16}$$

$$\sum_{i} \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p^*_{x_i|e_i}} - \mu(\theta) = \lambda,$$

$$\int r^*(\theta) d\theta = 1, \tag{17}$$

$$r^*(\theta) \geq 0,$$

$$\mu(\theta) \geq 0,$$

$$\mu(\theta) r^*(\theta) = 0\cdot$$

By summing Equation 15 over x_i and using Equation 17, we obtain

$$\lambda(e_i) = \int \pi(e_i \mid \theta) r^*(\theta) d\theta\cdot$$

Then, we can combine Equation 15 and 16 as follows:

$$p^*_{x_i|e_i} = \frac{\int \pi(x_i, e_i \mid \theta) r^*(\theta) d\theta}{\int \pi(e_i \mid \theta) r^*(\theta) d\theta} = R^*(x_i \mid e_i)\cdot$$

Equations 15 and 17 are the KKT conditions for the minimization problem in 13. Therefore, by assuming r is a probability distribution, these conditions provide necessary and sufficient conditions for a point to be a minimizer of $f(\cdot, \tilde{r})$ for a fixed $\tilde{r}$. Equation 12 combines these two equations and the assumption that r is a distribution into a single expression. □

Next, we will establish the connection between the saddle point and the central distribution. We also show that there is a strong relationship between central distributions and Bernardo's reference priors [3].

One significant insight provided by the next theorem is that it suggests inference and searching for a central distribution can be combined and performed simultaneously by finding a saddle point for f.

Definition 2.8 (Coverage)**.** Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model, and $\mathcal{X}$ be a query set. Let p be a prior distribution, and $P \in \mathcal{P}$ be the probability measure induced by p. We define the *coverage of p for $\mathcal{X}$* as

$$\mathbb{C}_\mathcal{X}[p] = \int \mathbb{D}_\mathcal{X}[P \mid \theta]\, p(\theta) d\theta\cdot$$

The coverage $\mathbb{C}_\mathcal{X}[p]$ is the dual of the divergence $\mathbb{D}_\mathcal{X}[P]$. For a query set, as we decrease the divergence, the coverage increases.

Theorem 3. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a query set. Consider the function f introduced in Lemma 2. If f has a saddle point, denoted by $(\mathbf{P}^*_\mathcal{X}, r^*)$, then R^*, the probability measure induced by r^*, is a central distribution for $\mathcal{X}$. Moreover, r^* has the maximum coverage in $\mathcal{M}_\Theta$.*

Proof. Suppose $C_\mathcal{X}$ is a central distribution for $\mathcal{X}$, and $(\mathbf{P}^*_\mathcal{X}, r^*)$ is a saddle point for f. Clearly, the point $(\mathbf{C}_\mathcal{X}, r^*)$, where $\mathbf{C}_\mathcal{X}$ is the vector of $C_\mathcal{X}(x_i \mid e_i)$ probabilities, is in the domain of f. Since $(\mathbf{P}^*_\mathcal{X}, r^*)$ is a saddle point for f, we have

$$f(\mathbf{P}^*_\mathcal{X}, r^*) \leq f(\mathbf{C}_\mathcal{X}, r^*)\cdot \tag{18}$$

Notice that the definition of f implies:

$$f(\mathbf{C}_\mathcal{X}, r^*) = \int \mathbb{D}_\mathcal{X}[C_\mathcal{X} \mid \theta]\, r^*(\theta) d\theta\cdot$$

Because r^* is a probability distribution and integrates to one we have

$$\int \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right] r^*(\theta) d\theta \leq \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right] = \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right],$$

and

$$f(\mathbf{P}^*_{\mathcal{X}}, r^*) \leq \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right] \cdot$$

$\mathcal{M}_{\Theta}$ is a generic parametric model, and therefore $R^* \in \mathcal{P}$, where R^* is the probability measure induced by r^*. Given that $(\mathbf{P}^*_{\mathcal{X}}, r^*)$ is a saddle point for f, by Lemma 2, the entries of $\mathbf{P}^*_{\mathcal{X}}$ satisfy Equation 9. Consequently, for the divergence measure of R^*, we can write

$$\begin{aligned}
\mathbb{D}_{\mathcal{X}}\left[R^*\right] &= \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}\left[R^* \mid \theta\right] \\
&= \max_{r} \int \mathbb{D}_{\mathcal{X}}\left[R^* \mid \theta\right] r(\theta) d\theta \\
&= \max_{r} f(\mathbf{P}^*_{\mathcal{X}}, r) \\
&= f(\mathbf{P}^*_{\mathcal{X}}, r^*),
\end{aligned}$$

where r denotes a probability distribution. Thus, Inequality 18 implies

$$\mathbb{D}_{\mathcal{X}}\left[R^*\right] \leq \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right] \cdot$$

Since $R^* \in \mathcal{P}$, we must have $\mathbb{D}_{\mathcal{X}}\left[R^*\right] = \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right]$, and R^* is a central distribution for $\mathcal{X}$.

Let q be an arbitrary prior distribution, and Q be the induced probability measure by q. Let $\mathbf{Q}_{\mathcal{X}}$ represent the vector of $Q(x_i \mid e_i)$ probabilities. By the definition of f, we have

$$\mathbb{C}_{\mathcal{X}}\left[q\right] = f(\mathbf{Q}_{\mathcal{X}}, q) \cdot$$

The point $(\mathbf{Q}_{\mathcal{X}}, q)$ satisfies Equation 12, and by Lemma 2, minimizes $f(\cdot, q)$. Therefore,

$$f(\mathbf{Q}_{\mathcal{X}}, q) \leq f(\mathbf{P}^*_{\mathcal{X}}, q) \cdot$$

On the other hand, $(\mathbf{P}^*_{\mathcal{X}}, r^*)$ is a saddle point for f, and

$$f(\mathbf{P}^*_{\mathcal{X}}, q) \leq f(\mathbf{P}^*_{\mathcal{X}}, r^*) \cdot$$

By Lemma 2, any saddle point for f, satisfies Equation 9. That implies

$$\mathbb{C}_{\mathcal{X}}\left[r^*\right] = f(\mathbf{P}^*_{\mathcal{X}}, r^*) \cdot$$

Thus, we have

$$\mathbb{C}_{\mathcal{X}}\left[q\right] \leq \mathbb{C}_{\mathcal{X}}\left[r^*\right],$$

and r^* has the maximum coverage. $\square$

Back to our dice example, we continue from Equation 5, and calculate the coverage of a prior distribution p for the query set $\mathcal{X} = \{X\}$, where X is the sample space of n dice rolls:

$$\begin{aligned}
\mathbb{C}_{\mathcal{X}}[p] &= \int \mathbb{D}_{\mathcal{X}}[P \mid \theta]\, p(\theta) d\theta \\
&= \int \sum_{x_1,\ldots,x_n} \pi(x_1,\ldots,x_n \mid \theta) \ln \frac{\pi(x_1,\ldots,x_n \mid \theta)}{P(x_1,\ldots,x_n)} p(\theta) d\theta \\
&= \int \sum_{x_1,\ldots,x_n} p(x_1,\ldots,x_n,\theta) \ln \frac{p(\theta \mid x_1,\ldots,x_n)}{p(\theta)} d\theta \cdot
\end{aligned}$$

We see that the central distribution for our dice example, when we consider a query set equivalent to the sample space, is the probability measure that is obtained by using the reference prior [3] for the multinomial parameters.

With the next theorem, we finally see why we refer to these distributions as "central." Exactly like the center of a circle, the central distribution is equidistant from the boundary.

Theorem 4 (Centrality). *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a query set. Suppose $C_{\mathcal{X}} \in \mathcal{P}$ is a probability measure induced by a strictly positive prior distribution. If there is a constant λ such that*

$$\mathbb{D}_{\mathcal{X}}[C_{\mathcal{X}} \mid \theta] = \lambda,$$

then $C_{\mathcal{X}}$ is a central distribution for $\mathcal{X}$.

Proof. Lemma 2 provides a set of sufficient conditions for a saddle point of f. Note that more restrictive conditions can also be imposed to derive an alternative set of sufficient conditions for a saddle point. Consequently, we can replace Inequality 10 with the more restrictive $r^*(\theta) > 0$ condition. By Equation 11, that implies $\mu(\theta) = 0$, and Lemma 2 conditions can be simplified to

$$\begin{gathered}
\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p^*_{x_i|e_i}} = \lambda, \\
p^*_{x_i|e_i} = R^*(x_i \mid e_i), \\
\int r^*(\theta) d\theta = 1, \quad r^*(\theta) > 0 \cdot
\end{gathered}$$

Let $C_{\mathcal{X}} \in \mathcal{P}$ be a probability measure, induced by a strictly positive prior $c_{\mathcal{X}}$. If for some constant λ, we have $\mathbb{D}_{\mathcal{X}}[C_{\mathcal{X}} \mid \theta] = \lambda$, we see that the point

$(\mathbf{C}_{\mathcal{X}}, c_{\mathcal{X}})$ satisfies these simplified conditions, where $\mathbf{C}_{\mathcal{X}}$ is the corresponding vector of $C_{\mathcal{X}}(x_i \mid e_i)$ probabilities.

Therefore, Lemma 2 implies that $(\mathbf{C}_{\mathcal{X}}, c_{\mathcal{X}})$ is a saddle point for f. Since $\mathcal{M}_\Theta$ is a generic parametric model, Theorem 3 implies $C_{\mathcal{X}}$ is a central distribution for $\mathcal{X}$. □

In the dice example, following the principle of objectivity led us to define a query set that included vectors representing n trials of our experiment. This query set contained two variables, X and Y: X represented the sample space, while Y corresponded to trials of a two-sided dice indicating whether the original dice result was odd or even.

In many cases, we consider a query set structured as follows:

$$\mathcal{X} = \{X, X_1, X_2, \dots, X_k\},$$

where X represents the whole sample space, and each X_i represents n trials of a transformed experiment. We usually assume a bounded, but sufficiently large number of trials. The transformed experiments are functions of the original experiment and can be parametrized by functions of the model parameter. In the next theorem, for query sets of this form, we will develop a fixed point equation characterizing the central distribution.

Theorem 5. *Suppose $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a generic parametric model, and $\mathcal{X} = \{X, X_1, X_2, \dots, X_k\}$ is a query set in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X_i, That is, for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x_i \mid \theta) = \pi\left(x_i \mid \omega_i\right) \cdot \tag{19}$$

Also, let $\Omega(\theta) = (\omega_1, \omega_2, \dots, \omega_k) = \omega$.

We define a map, denoted by T, on the space of unnormalized probability measures of ω, as follows:

$$T\left[\rho\right](\omega) = \left(\frac{\varphi_\rho(\omega)}{\rho(\omega)} \prod_i \frac{\varphi_\rho(\omega_i)}{\rho(\omega_i)}\right)^{\frac{1}{k+1}} \rho(\omega), \tag{20}$$

where ρ is an unnormalized probability measure, and:

$$\varphi_\rho(\omega_i) = \exp\left\{\sum_{x_i} \pi(x_i \mid \omega_i) \ln \rho(\omega_i \mid x_i)\right\},$$

$$\varphi_\rho(\omega) = \int_{\{\theta:\Omega(\theta)=\omega\}} \varphi_\rho(\theta) d\theta,$$

$$\varphi_\rho(\theta) = \exp\left\{\sum_{x} \pi(x \mid \theta) \ln \rho(\theta \mid x)\right\}\cdot$$

Let r be a strictly positive prior probability distribution. The unnormalized measure μr is a fixed point for T, and

$$r(\theta \mid \omega) = \frac{\varphi_r(\theta)}{\varphi_r(\omega)}, \tag{21}$$

if and only if

$$\mathbb{D}_{\mathcal{X}}\left[R \mid \theta\right] = (k+1)\ln \mu, \tag{22}$$

where R is the probability measure induced by r.

Proposition. We have

$$\mathbb{C}_{\mathcal{X}}\left[r\right] = (k+1)\ln \mu\cdot$$

Proof. To begin, consider that for any arbitrary probability distribution P, induced by a strictly positive prior p, using Equation 19, we can derive the following:

$$\begin{aligned}
\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] &= \sum_x \pi(x \mid \theta) \ln \frac{\pi(x \mid \theta)}{P(x)} + \sum_i \sum_{x_i} \pi(x_i \mid \theta) \ln \frac{\pi(x_i \mid \theta)}{P(x_i)} \\
&= \sum_x \pi(x \mid \theta) \ln \frac{p(\theta \mid x)}{p(\theta)} + \sum_i \sum_{x_i} \pi(x_i \mid \omega_i) \ln \frac{p(\omega_i \mid x_i)}{p(\omega_i)} \\
&= \ln \frac{\varphi_p(\theta)}{p(\theta)} + \sum_i \ln \frac{\varphi_p(\omega_i)}{p(\omega_i)} \\
&= \ln \left(\frac{\varphi_p(\theta)}{p(\theta)} \prod_i \frac{\varphi_p(\omega_i)}{p(\omega_i)}\right)\cdot
\end{aligned}$$

Since $\Omega(\theta) = \omega$, we have $p(\theta) = p(\theta \mid \omega)p(\omega)$, and

$$\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] + \ln p(\theta \mid \omega) = \ln \left(\frac{\varphi_p(\theta)}{p(\omega)} \prod_i \frac{\varphi_p(\omega_i)}{p(\omega_i)}\right)\cdot \tag{23}$$

To prove the 'if' direction, we assume for every value of θ, Equation 22 holds for a probability measure R, induced by a strictly positive prior r. We exponentiate both sides of Equation 23 to obtain

$$\mu^{k+1} r(\theta \mid \omega) = \frac{\varphi_r(\theta)}{r(\omega)} \prod_{i=1}^{k} \frac{\varphi_r(\omega_i)}{r(\omega_i)}. \tag{24}$$

Since this equation holds for any value of θ, we can integrate both sides with respect to θ:

$$\int_{\{\theta:\Omega(\theta)=\omega\}} \mu^{k+1} r(\theta \mid \omega) d\theta = \int_{\{\theta:\Omega(\theta)=\omega\}} \left(\frac{\varphi_r(\theta)}{r(\omega)} \prod_i \frac{\varphi_r(\omega_i)}{r(\omega_i)} \right) d\theta.$$

That implies

$$\mu^{k+1} = \left(\frac{\varphi_r(\omega)}{r(\omega)} \prod_i \frac{\varphi_r(\omega_i)}{r(\omega_i)} \right). \tag{25}$$

Now we evaluate T at μr:

$$\begin{aligned} T\left[\mu r\right](\omega) &= \left(\frac{\varphi_{\mu r}(\omega)}{\mu r(\omega)} \prod_i \frac{\varphi_{\mu r}(\omega_i)}{\mu r(\omega_i)} \right)^{\frac{1}{k+1}} \mu r(\omega) \\ &= \left(\mu^{k+1}\right)^{\frac{1}{k+1}} r(\omega) \\ &= \mu r(\omega). \end{aligned}$$

Therefore, μr is a fixed point for T. In addition, for $r(\theta \mid \omega)$ we can obtain Equation 21 by dividing Equation 24 by Equation 25. This completes the proof of the forward direction.

We now prove the 'only if' direction. Suppose μr is a fixed point of T, where r is a strictly positive probability distribution and μ is a constant. Since $r(\omega) \neq 0$, by letting $T\left[\mu r\right] = \mu r$, for every value of ω we get

$$\left(\frac{\varphi_r(\omega)}{r(\omega)} \prod_i \frac{\varphi_r(\omega_i)}{r(\omega_i)} \right)^{\frac{1}{k+1}} = \mu. \tag{26}$$

On the other hand, under the assumptions of the theorem, Equation 21 holds. Substituting this into Equation 23 gives

$$\mathbb{D}_{\mathcal{X}}\left[R \mid \theta\right] = \ln \left(\frac{\varphi_r(\omega)}{r(\omega)} \prod_i \frac{\varphi_r(\omega_i)}{r(\omega_i)} \right).$$

Now, substituting Equation 26 into the preceding equation yields

$$\mathbb{D}_{\mathcal{X}}\left[R \mid \theta\right] = (k+1)\ln\mu,$$

which concludes the proof. $\square$

Using Equation 20, we can construct a sequence of unnormalized measures by

$$\rho_{n+1} = T\left[\rho_n\right] \cdot$$

This sequence can be used in an iterative algorithm to find an estimation of the central prior $c_{\mathcal{X}}(\omega)$. Once $c_{\mathcal{X}}(\omega)$ is determined, Equation 21 can be used to obtain $c_{\mathcal{X}}(\theta)$.

The key property of the mapping T is that it is defined on the space of unnormalized measures. This allows us to compute $c_{\mathcal{X}}(\theta)$ for specific regions of the parameter space without needing to scan the entire space, which is typically large and intractable. Additionally, it works with ω and its marginals ω_i, rather than θ. Notably, the space of ω is significantly smaller than that of θ.

In Equation 20, if we replace φ_ρ with $\varphi_{c_{\mathcal{X}}}$, it can improve the convergence properties considerably. In models that the posterior of the parameter converges, we can calculate $\varphi_{c_{\mathcal{X}}}$ by using any well behaved prior instead of $c_{\mathcal{X}}$. We typically use a conjugate prior to simplify the derivation of the posterior potential. By using a conjugate prior, the posterior potential takes a form similar to the entropy of the likelihood distribution.

Definition 2.9 (Posterior Potential)**.** Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model, and X be a random variable defined on $\mathcal{S}$. Let Ω be a function of the parameter Θ that fully parametrizes X.

If the unnormalized measure $\varphi(\omega)$, defined as

$$\varphi(\omega) = \exp\left\{\sum_x \pi(x \mid \omega)\ln p(\omega \mid x)\right\},$$

dose not depend on $p(\omega)$, we refer to it as the *posterior potential* of ω. *(This is not formal enough)*

Theorem 6. *Let T be the map defined as*

$$T\left[\rho\right](\omega) = \left(\frac{\varphi(\omega)}{\rho(\omega)}\prod_i \frac{\varphi(\omega_i)}{\rho(\omega_i)}\right)^{\frac{1}{k+1}} \rho(\omega),$$

where $\varphi(\cdot)$ denotes the posterior potential. The sequence

$$\rho_{n+1} = T\left[\rho_n\right],$$

converges to the fixed point of T, for any arbitrary initial unnormalized probability measure ρ_1.

Proof. We prove the theorem by using Banach fixed point theorem [1]. Let ρ and η be two unnormalized probability measures of a random variable ω We define the distance of ρ and η as follows:

$$d_\Omega\left(\rho,\eta\right) = \int \left|\ln\frac{\rho(\omega)}{\eta(\omega)}\right| d\omega\cdot \tag{27}$$

Obviously, if $\rho = \eta$, then $d_\Omega\left(\rho,\eta\right) = 0$. Conversely, if $d_\Omega\left(\rho,\eta\right) = 0$, that implies $\rho = \eta$ because $\int \left|f(\omega)\right| d\omega = 0$ if and only if $f(\omega) = 0$. Moreover, d_Ω is symmetric $d_\Omega\left(\rho,\eta\right) = d_\Omega\left(\eta,\rho\right)$, and satisfies the triangle inequality:

$$\begin{aligned}
d_\Omega\left(\rho,\phi\right) + d_\Omega\left(\phi,\eta\right) &= \int \left|\ln\frac{\rho(\omega)}{\phi(\omega)}\right| d\omega + \int \left|\ln\frac{\phi(\omega)}{\eta(\omega)}\right| d\omega \\
&= \int \left(\left|\ln\frac{\rho(\omega)}{\phi(\omega)}\right| + \left|\ln\frac{\phi(\omega)}{\eta(\omega)}\right|\right) d\omega \\
&\geq \int \left|\ln\frac{\rho(\omega)}{\phi(\omega)} + \ln\frac{\phi(\omega)}{\eta(\omega)}\right| d\omega \\
&= d_\Omega\left(\rho,\eta\right)\cdot
\end{aligned}$$

Thus, d_Ω is a metric.

Now, we show that T is a contraction.

$$\begin{aligned}
d_\Omega\left(T\left[\rho\right], T\left[\eta\right]\right) &= \int \left|\ln\frac{T\left[\rho\right](\omega)}{T\left[\eta\right](\omega)}\right| d\omega \\
&= \int \left|\ln\left(\frac{\rho(\omega)}{\eta(\omega)}\left(\frac{\eta(\omega)}{\rho(\omega)}\prod_i \frac{\eta(\omega_i)}{\rho(\omega_i)}\right)^{\frac{1}{k+1}}\right)\right| d\omega \\
&= \frac{1}{k+1}\int \left|\ln\left(\prod_i \frac{p_\rho(\omega \mid \omega_i)}{p_\eta(\omega \mid \omega_i)}\right)\right| d\omega \\
&\leq \frac{1}{k+1}\sum_{i=1}^{k}\int \left|\ln\frac{p_\rho(\omega \mid \omega_i)}{p_\eta(\omega \mid \omega_i)}\right| d\omega \\
&\leq^{?} \frac{k}{k+1} d_\Omega\left(\rho,\eta\right)
\end{aligned}$$

Unfortunately, this proof is not complete. □

2.3 Multinomial Model

In this section, we examine the derivation of central distributions in multinomial models. Multinomial models play a key role in statistical learning. By de Finetti's theorem, any exchangeable sequence of discrete random variables can be represented by a multinomial model. Exchangeable sequences offer a framework for modeling data with independence and order invariance, which is particularly important in probabilistic and statistical applications.

A multinomial model is a parametric model (Definition 1.2) in which the likelihood function follows a multinomial distribution. The configuration of the model is determined by two variables: n and d. Each element of the sample space $s \in \mathcal{S}$ can be represented by a d dimensional random vector

$$X = (x_1, x_2, \ldots, x_d),$$

where each x_i is a nonnegative integer such that $\sum_{i=1}^{d} x_i = n$. The model parameter is a nonnegative d dimensional real vector

$$\theta = (\theta_1, \theta_2, \ldots, \theta_d),$$

where $\sum_{i=1}^{d} \theta_i = 1$. The likelihood distributing $\pi(X \mid \theta)$ is a multinomial distribution

$$\pi(x \mid \theta) = \frac{n!}{\prod_{i=1}^{d} x_i!} \prod_{i=1}^{d} \theta_i^{x_i}.$$

In a multinomial model, as n increases, the posterior distribution of θ converges to the same distribution regardless of the prior. Therefore, when n is sufficiently large, we can use a uniform prior to obtain the posterior potential of the multinomial parameter. The uniform prior is conjugate to the multinomial distribution, and the resulting posterior will follow a Dirichlet distribution:

$$\begin{aligned} p(\theta \mid x) &\sim \mathrm{Dir}(x_1 + 1, x_2 + 1, \ldots, x_d + 1) \\ &= \prod_{j=1}^{d-1} (n + j) \left(\frac{n!}{\prod_{i=1}^{d} x_i!} \prod_{i=1}^{d} \theta_i^{x_i} \right). \end{aligned}$$

This implies that for the posterior potential we have

$$\begin{aligned} \varphi(\theta) &= \exp \left\{ \sum_{x} \pi(x \mid \theta) \ln p(\theta \mid x) \right\} \\ &= e^{-\mathbb{H}[X \mid \theta]} \prod_{j=1}^{d-1} (n + j), \end{aligned}$$

where $\mathbb{H}[X \mid \theta]$ is the entropy of the multinomial distribution with parameter θ. The entropy of the multinomial distribution can be written in the following form:

$$\mathbb{H}[X \mid \theta] = -\ln(n!) - \sum_{i=1}^{d} \ln \phi(\theta_i),$$

where

$$\phi(\theta_i) = \exp\left\{ n\theta_i \ln \theta_i - \sum_{j=0}^{n} \ln(j!) \binom{n}{j} \theta_i^j (1-\theta_i)^{n-j} \right\} \cdot \tag{28}$$

Therefore, for the posterior potential of the multinomial parameter we have

$$\varphi(\theta) = (n+d-1)! \prod_{i=1}^{d} \phi(\theta_i) \cdot \tag{29}$$

When using a multinomial model, we usually consider query sets that consist of random variables corresponding to fusions of the multinomial categories.

Let $X = (x_1, x_2, \ldots, x_d)$ be a random vector representing an element of the sample space of a multinomial model. A fusion random variable Y is obtained by summing up some entries of X. More precisely, if $\mathcal{A}_1, \ldots, \mathcal{A}_{d'}$ is a partition of the set $\{1, 2, \ldots, d\}$, each value of Y is a vector $(y_1, y_2, \ldots, y_{d'})$ where

$$y_j = \sum_{i \in \mathcal{A}_j} x_i, \qquad j = 1, \ldots, d' \cdot$$

The fusion variable Y also follows a multinomial distribution with the parameter $\omega = (w_1, w_2, \ldots, w_{d'})$, where

$$w_j = \sum_{i \in \mathcal{A}_j} \theta_i, \qquad j = 1, \ldots, d' \cdot$$

For example $(\theta_1+\theta_5, \theta_2, \theta_4+\theta_7+\theta_8, \ldots)$ fully parametrizes the fusion variable $(x_1 + x_5, x_2, x_4 + x_7 + x_8, \ldots)$.

Suppose that in a multinomial model we aim to find the central distribution for a query set $\mathcal{X} = \{X, Y_1, Y_2, \ldots, Y_k\}$, where each Y_i is a fusion variable, and X represents the original sample space. For using theorem 5, we need to obtain $\varphi(\omega_i)$, where ω_i is the parameter of Y_i, and also $\varphi(\omega)$, where $\omega = (\omega_1, \omega_2, \ldots, \omega_k)$. Note that each ω_i is a multinomial parameter.

Each $\varphi(\omega_i)$ can be easily obtained using Equation 29. However, calculating $\varphi(\omega)$ requires integration:

$$\varphi(\omega) = \int_{\{\theta : \Omega(\theta) = \omega\}} \varphi(\theta) d\theta \cdot$$

This integration is performed over the set of all nonnegative solutions to the linear system of equations $\Omega(\theta) = \omega$. Here, Ω is a linear map with a matrix M_Ω consisting solely of 0 and 1 entries. The range of Ω typically has much lower dimensionality than θ, and the system of equations contains many free variables. This equation system has a solution when ω is inside the convex hull of the columns of M_Ω.

We convert this integration into an inference problem in a probabilistic graphical model, and then use a slightly modified version of the clique tree message passing [5] algorithm.

For representing the equations of the linear system $\Omega(\theta) = \omega$ by a graphical model, we use indicator factors. Let $\vartheta \subseteq \{\theta_1, \theta_2, \ldots, \theta_d\}$, and $0 \le w \le 1$. An indicator factor ξ is a function with the scope $\{\vartheta, w\}$ such that:

$$\xi(\vartheta, w) = \begin{cases} 1 & \sum_{\theta_i \in \vartheta} \theta_i = w, \\ 0 & \text{otherwise}\cdot \end{cases}$$

For example $\xi(\theta_2, \theta_5, \theta_6, w)$ is 1 when $\theta_2 + \theta_5 + \theta_6 = w$, and 0 otherwise.

Suppose the rank of M_Ω is r. We select r linearly independent equations from $\Omega(\theta) = \omega$, and define an indicator factor for every equation. Then, we construct an unnormalized measure ψ:

$$\psi(\theta, w_1, w_2, \ldots, w_r) = \prod_{i=1}^{d} \phi(\theta_i) \prod_{k=1}^{r} \xi(\vartheta_k, w_k)\cdot \tag{30}$$

Clearly, marginalizing ψ over θ yields $\varphi(\omega)$. To achieve this, we construct a clique tree for ψ, denoted by $\mathcal{T}_\psi$, that has at least one clique C_w, containing all w_k for $k \in \{1, 2, \ldots, r\}$. By performing message passing and selecting C_w as the root of the tree, we can compute the desired marginal $\psi(w_1, \ldots, w_r)$ after summing out other variables in C_w.

In $\mathcal{T}_\psi$ each equation and each variable of the equation system $\Omega(\theta) = \omega$ is associated with a clique. Let C_i and C_j be two adjacent cliques in $\mathcal{T}_\psi$ with the sepset $\mathcal{S}_{i,j}$. Without loss of generality, we assume that the root C_w, is on the C_j side of the tree. Let $\mathcal{W}_{<(i,j)}$ be the set of all w_k for $k \in \{1, 2, \ldots, r\}$, such that an indicator $\xi(\cdot, w_k)$ is associated with a clique on the C_i side of the tree. We can think of $\mathcal{W}_{<(i,j)}$ as the set of all equations that are associated with the C_i side of the tree. Let $\vartheta_{i,j}$ be the set of all θ_i for $i \in \{1, 2, \ldots, d\}$ (i.e. the variables of the equation system) that are on the scope of both C_i and C_j. Due to the running intersection property, we have

$$\mathcal{W}_{<(i,j)} \cup \vartheta_{i,j} \subseteq \mathcal{S}_{i,j}\cdot$$

However, during message passing, we can use a simplified sepset instead.

Definition 2.10. Let $\mathcal{T}$ be a clique tree. Let $\mathcal{R}_{i,j}$ denote the set of all possible assignments to the sepset $\mathcal{S}_{i,j}$. A *message reparameterization* for $\mathcal{T}$ is a map (i.e. function) $f_{i,j}$ defined from $\mathcal{R}_{i,j}$ to another set $\mathcal{R}^*_{i,j}$ for every i, j, which is message preserving. That means, for any $s_{i,j}, s'_{i,j} \in \mathcal{R}_{i,j}$, if $f_{i,j}(s_{i,j}) = f_{i,j}(s'_{i,j})$ then $\delta_{i\to j}(s_{i,j}) = \delta_{i\to j}(s'_{i,j})$, where $\delta_{i\to j}$ denotes the message from C_i to C_j.

The reparameterization of $\delta_{i\to j}$, is a function $\sigma_{i\to j}$ defined on $\mathcal{R}^*_{i,j}$ such that for every $s_{i,j} \in \mathcal{R}_{i,j}$, we have $\delta_{i\to j}(s_{i,j}) = \sigma_{i\to j}(f_{i,j}(s_{i,j}))$.

Proposition. We can replace all messages $\delta_{i\to j}$ with their reparameterization $\sigma_{i\to j}$ without affecting the message passing algorithm.

Proof. During message passing no information regarding specific assignments to $\mathcal{S}_{i,j}$ is passed between cliques. C_i simply produces $\delta_{i\to j}$ for every possible assignment to $\mathcal{S}_{i,j}$. Let Nb_i be the set of indexes of cliques that are neighbors of C_i. Having the mappings $f_{i,j}$ and $f_{k,i}$ for every $k \in Nb_i$, the reparameterized messages $\sigma_{k\to i}$ can be used to produce $\sigma_{i\to j}$. $\square$

Suppose $w_k \in \mathcal{W}_{<(i,j)}$ and $\xi(\vartheta_k, w_k)$ is the indicator factor containing w_k. This indicator represents the equation

$$\sum_{\theta_i \in \vartheta_k} \theta_i = w_k.$$

We define a new variable $0 \le w_k^{(i,j)} \le 1$ for every $w_k \in \mathcal{W}_{<(i,j)}$ such that

$$w_k^{(i,j)} = w_k - \sum_{\theta_i \in \vartheta_k \cap \vartheta_{i,j}} \theta_i.$$

For each sepset $\mathcal{S}_{i,j}$, there will be a corresponding set of these new variables. We denote this set by $\mathcal{W}^*_{i,j}$. We have a map from assignments to $\mathcal{S}_{i,j}$ into assignments to $\mathcal{W}^*_{i,j}$, which is message preserving. If two assignments $s_{i,j}$, $s'_{i,j}$ to the sepset variables induce the same assignment to $\mathcal{W}^*_{i,j}$ variables, that implies the equation system corresponding to the C_i part of the tree has the same solution set for $s_{i,j}$ and $s'_{i,j}$. As a result, we will have $\delta_{i\to j}(s_{i,j}) = \delta_{i\to j}(s'_{i,j})$.

We can simplify message computation by introducing certain constraints on the structure of $\mathcal{T}_\psi$.

Definition 2.11 (Convolution Clique Tree)**.** Let $\mathcal{T}_\psi$ be a clique tree for the factorized measure of Equation 30. Let $pr(i)$, for each clique C_i, be the index of the parent of C_i, i.e. the neighbor on the path to the root clique C_w. We refer to $\mathcal{T}_\psi$ as a *convolution clique tree* if for every i we have

$$C_i - \mathcal{S}_{i,pr(i)} = \{\theta_i\}.$$

That is, in each clique exactly one variable is eliminated. We also require that each $\phi(\theta_i)$ factor be associated with the clique that eliminates θ_i.

Note that indicator factors may be associated with any clique as long as the tree remains family preserving.

Definition 2.12. In a convolution clique tree, a clique that is associated with at least one indicator factor is an *assignment node*. Any clique that is not an assignment node is a *convolution node*.

To make the notation more convenient, we can express messages as functions of vectors. Thus, the reparameterized message $\sigma_{i\to j}$ will be a function of a vector $w^*_{i,j}$, with an entry for each variable in $\mathcal{W}^*_{i,j}$. Note that each variable in $\mathcal{W}^*_{i,j}$ corresponds to an equation. Alternatively, $\mathcal{W}^*_{i,j}$ can be viewed as a set of equations. We define $\mathbf{1}_{i,j}$, for each i, j, to be an indicator vector indicating those equations of $\mathcal{W}^*_{i,j}$ that include θ_i (the variable that is elaminated in C_i).

If C_i is a convolution node, the outgoing message from C_i to C_j can be computed by convolution:

$$\sigma_{i\to j}(w^*_{i,j}) = \int \phi(t) \prod_{k\in Nb_i - \{j\}} \sigma_{k\to i}(w^*_{k,i} - t\mathbf{1}_{k,i})dt$$

On the other hand, when C_i is an assignment node, for every value of $w^*_{i,j}$ there is only one acceptable value for θ_i, and the message can be computed without convolution. Moreover, certain values of $w^*_{i,j}$ will be invalid. For each indicator factor associated with C_i that does not include θ_i, the corresponding variable in $\mathcal{W}^*_{i,j}$ must always be 0 in the computed message. For those that include θ_i, the corresponding variables in $\mathcal{W}^*_{i,j}$ must be the same value and equal. If we denote this value by t, we have

$$\sigma_{i\to j}(w^*_{i,j}) = \phi(t) \prod_{k\in Nb_i - \{j\}} \sigma_{k\to i}(w^*_{k,i} - t\mathbf{1}_{k,i})\cdot$$

In the above equation, the entries of $w^*_{i,j}$ that correspond to equations including θ_i are t, the entries corresponding to equations that do not include θ_i are 0, and other entries can take any value.

A Expectation as a Linear Map

In this section, we present a fresh perspective on probability by introducing an alternative axiomatization of probability theory. This formulation highlights the deep connections between probability theory and linear algebra.

Definition A.1 (Sample Space)**.** A sample space, denoted by $\mathcal{S}$, is the set of all possible outcomes of a random experiment. That is, the outcome of a random experiment can *always* be represented by *exactly one* element of $\mathcal{S}$.

Definition A.2 (Action Space)**.** Let $\mathcal{S}$ be a sample space. The action space of $\mathcal{S}$, denoted by $\mathcal{A}_{\mathcal{S}}$, is the set of all functions from $\mathcal{S}$ to $\mathbb{R}$, where $\mathbb{R}$ is the field of real numbers.

Proposition. The action space equals $\mathbb{R}^{\mathcal{S}}$, and is a vector space.

We can think of the members of $\mathcal{A}_{\mathcal{S}}$ as hypothetical actions or gambles. Each action associates an amount of loss with every possible outcome $s \in \mathcal{S}$. This loss is quantified by a real number. A negative loss indicates profit.

Definition A.3 (Sure Thing)**.** The *sure thing*, denoted by c, is an element of $\mathcal{A}_{\mathcal{S}}$, such that $c(s) = 1$ for every $s \in \mathcal{S}$.

Definition A.4. Let $\mathcal{E} \subseteq \mathcal{S}$. The restriction to $\mathcal{E}$ operator, denoted by $R_{\cap\mathcal{E}}$, is an operator on $\mathcal{A}_{\mathcal{S}}$, such that $\ell' = R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})$ for any $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$ is an element of $\mathcal{A}_{\mathcal{S}}$ and

$$\ell'(s) = \begin{cases} \ell_{\mathcal{S}}(s) & s \in \mathcal{E}, \\ 0 & s \in \mathcal{S} - \mathcal{E}. \end{cases}$$

We denote the range of $R_{\cap\mathcal{E}}$ by $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$.

Proposition. The restriction operator is a linear map.

Proposition. $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ is an invariant subspace of $\mathcal{A}_{\mathcal{S}}$ under $R_{\cap\mathcal{E}}$. If $\{\mathcal{E}_1, \mathcal{E}_2, \ldots\}$ is a partition of $\mathcal{S}$, then

$$\mathcal{A}_{\mathcal{S}} = \mathcal{A}_{\mathcal{S}\cap\mathcal{E}_1} \oplus \mathcal{A}_{\mathcal{S}\cap\mathcal{E}_2} \oplus \ldots,$$

where $\oplus$ indicates direct sum [2].

The restriction operator restricts an action to a subset of the sample space by simply ignoring its payoff for any outcome that is not in that subset.

Definition A.5 (Linear Functional)**.** Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$. A linear functional on $\mathcal{V}$ is a linear map from $\mathcal{V}$ to $\mathbb{F}$.

Definition A.6 (Expectation)**.** An *expectation* for $\mathcal{S}$ is a linear functional on $\mathcal{A}_{\mathcal{S}}$, denoted by $\mathbb{E}_{\mathcal{S}}$, with the following properties:

- For the sure thing $c_{\mathcal{S}}$, we have $\mathbb{E}_{\mathcal{S}}[c_{\mathcal{S}}] = 1$.
- For any $\mathcal{E} \subseteq \mathcal{S}$, we have $\mathbb{E}_{\mathcal{S}}[R_{\cap\mathcal{E}}(c_{\mathcal{S}})] \geq 0$.

Proposition. $\mathbb{E}_{\mathcal{S}}$ is an element of the dual space of $\mathcal{A}_{\mathcal{S}}$.

The expectation map quantifies our expectation about an action as a real number. It must have three properties: Most importantly it should be linear. That means if the payoff of an action doubles, our expectation from that action should be doubled too. If we add

Definition A.7 (Conditional Expectation)**.** Let $\mathbb{E}_{\mathcal{S}}$ be an expectation for $\mathcal{S}$ and let $\mathcal{E} \subseteq \mathcal{S}$. $\mathbb{E}_{\mathcal{S}}$ conditioned on $\mathcal{E}$, denoted by $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$, is a linear map on $\mathcal{S}$ with the following properties:

- For the sure thing $c_{\mathcal{S}}$, we have $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[c_{\mathcal{S}}\right] = 1$.
- For any $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$, and $\mathcal{E} \subseteq \mathcal{S}$, we have $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell_{\mathcal{S}}\right] = 0$ if and only if $\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})\right] = 0$.

At this point, we have a complete framework for decision-making under uncertainty. Interestingly, we do not need to explicitly define probability. However, in order to present a more familiar framework, we provide a definition of conventional probability.

Definition A.8 (Probability)**.** We define the probability of $\mathcal{E} \subseteq \mathcal{S}$ as follows:

$$P(\mathcal{E}) = \mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(c_{\mathcal{S}})\right] \cdot$$

Theorem 7. *Definition A.8 satisfies Kolmogorov's axioms of probability.*

We now will establish the familiar relationship between conditional expectation and normal expectation.

Lemma 3. *Let $\mathcal{F}$ and $\mathcal{G}$ be two linear functionals defined on a vector space $\mathcal{V}$. $\mathcal{F}$ and $\mathcal{G}$ have the same null space if and only if $\mathcal{F} = \lambda\mathcal{G}$ for some $\lambda \neq 0$.*

Proof. If $\mathcal{F} = \lambda\mathcal{G}$, $\mathcal{F}[v] = 0$ if and only if $\mathcal{G}[v] = 0$ for every $v \in \mathcal{V}$. This proves the if part.

We prove the only if part by contradiction. Assume $\mathcal{F} \neq \lambda\mathcal{G}$. Then there exist $u, v \in \mathcal{V}$ such that

$$\mathcal{F}[u]\,\mathcal{G}[v] - \mathcal{F}[v]\,\mathcal{G}[u] \neq 0 \cdot$$

Let $w = \mathcal{F}[u]\,v - \mathcal{F}[v]\,u$. Since $\mathcal{V}$ is a vector space $w \in \mathcal{V}$. $\mathcal{F}$ and $\mathcal{G}$ are linear. Hence, $\mathcal{F}[w] = 0$ and $\mathcal{G}[w] = \mathcal{F}[u]\,\mathcal{G}[v] - \mathcal{F}[v]\,\mathcal{G}[u] \neq 0$ and the null spaces of $\mathcal{F}$ and $\mathcal{G}$ are not equal. This contradiction completes the proof. $\square$

Theorem 8. *Let $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$ and $\mathcal{E} \subseteq \mathcal{S}$, where $\mathcal{S}$ is a sample space. If $R_{\cap\mathcal{E}}$ is the restriction operator to $\mathcal{E}$, then we have*

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell_{\mathcal{S}}\right] = \frac{\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})\right]}{P(\mathcal{E})}\cdot$$

Proof. For an arbitrary $\ell \in \mathcal{A}_{\mathcal{S}}$, let $\ell' = \ell - R_{\cap\mathcal{E}}(\ell)$. Clearly, $R_{\cap\mathcal{E}}(\ell')$ is the zero function, and therefore $\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell')\right] = 0$. Since $\mathcal{A}_{\mathcal{S}}$ is a vector space we have $\ell' \in \mathcal{A}_{\mathcal{S}}$, and the definition of conditional expectation implies $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell'\right] = 0$. Therefore

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell\right] - \mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[R_{\cap\mathcal{E}}(\ell)\right] = 0\cdot \tag{31}$$

Let $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ represent the range of $R_{\cap\mathcal{E}}$. $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ is a subspace of $\mathcal{A}_{\mathcal{S}}$. By the definition of conditional expectation, functionals obtained by the restrictions of $\mathbb{E}_{\mathcal{S}}$ and $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$ to $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ have the same null space. That implies by Lemma 3 for every $\ell \in \mathcal{A}_{\mathcal{S}}$ we have

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[R_{\cap\mathcal{E}}(\ell)\right] = \lambda\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell)\right]\cdot$$

Thus, by Equation 31

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell\right] = \lambda\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell)\right]\cdot$$

On the other hand, for the sure thing $c_{\mathcal{S}}$, we have

$$\lambda = \frac{\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[c_{\mathcal{S}}\right]}{\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(c_{\mathcal{S}})\right]} = \frac{1}{P(\mathcal{E})}\cdot$$

This proves the theorem. □

Definition A.9 (Dual Map)**.** Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. Suppose $M : \mathcal{A} \to \mathcal{B}$ is a map from $\mathcal{A}$ to $\mathcal{B}$. The dual map of M is the map $M^* : \mathbb{R}^{\mathcal{B}} \to \mathbb{R}^{\mathcal{A}}$, defined for each $f \in \mathbb{R}^{\mathcal{B}}$ by

$$M^*(f) = f \circ M\cdot$$

Proposition. The dual map is a linear map.

Theorem 9. *Let $\mathcal{S}$ be a sample space. Suppose $M : \mathcal{S} \to \mathcal{S}'$ is a map from $\mathcal{S}$ to another set $\mathcal{S}'$. If $\mathbb{E}_{\mathcal{S}}$ is an expectation for $\mathcal{S}$, then $\mathbb{E}_{\mathcal{S}'} = M^{**}(\mathbb{E}_{\mathcal{S}})$ is an expectation for $\mathcal{S}'$, where M^{**} is the dual map of M^*, and M^* is the dual map of M.*

Proof. M^{**} is a dual map and therefore is linear.

By definition for every $\ell_{\mathcal{S}'} \in \mathcal{A}_{\mathcal{S}'}$ we have

$$\mathbb{E}_{\mathcal{S}'}\left[\ell_{\mathcal{S}'}\right] = \mathbb{E}_{\mathcal{S}} \circ \ell_{\mathcal{S}'} \circ M\cdot$$

We can easily verify that

$$c_{\mathcal{S}'} \circ M = c_{\mathcal{S}}\cdot$$

Therefore, $\mathbb{E}_{\mathcal{S}'}\left[c_{\mathcal{S}'}\right] = 1$.

On the other hand for any $\mathcal{E}' \subseteq \mathcal{S}'$, we have

$$R_{\cap\mathcal{E}'}(c_{\mathcal{S}'}) \circ M = R_{\cap\mathcal{E}}(c_{\mathcal{S}}),$$

where

$$\mathcal{E} = \{s \in \mathcal{S} : M(s) \in \mathcal{E}'\}\cdot$$

This implies $\mathbb{E}_{\mathcal{S}'}\left[R_{\cap\mathcal{E}'}(c_{\mathcal{S}'})\right] \geq 0$. □

References

[1] Praveen Agarwal, Mohamed Jleli, and Bessem Samet. *Banach Contraction Principle and Applications*, pages 1–23. Springer Singapore, Singapore, 2018.

[2] Sheldon Jay Axler. *Linear Algebra Done Right*. Undergraduate Texts in Mathematics. Springer, New York, 2023.

[3] José M. Bernardo. Reference analysis. In D.K. Dey and C.R. Rao, editors, *Bayesian Thinking*, volume 25 of *Handbook of Statistics*, pages 17–90. Elsevier, 2005.

[4] Stephen Boyd and Lieven Vandenberghe. *Convex Optimization*. Cambridge University Press, 2004.

[5] D. Koller and N. Friedman. *Probabilistic Graphical Models: Principles and Techniques*. Adaptive computation and machine learning. MIT Press, 2009.